The coordination complex $[CoCl_2(en)_2]^+$ has three isomers, two optically active cis forms and one optically inactive trans form. These structures are shown below.

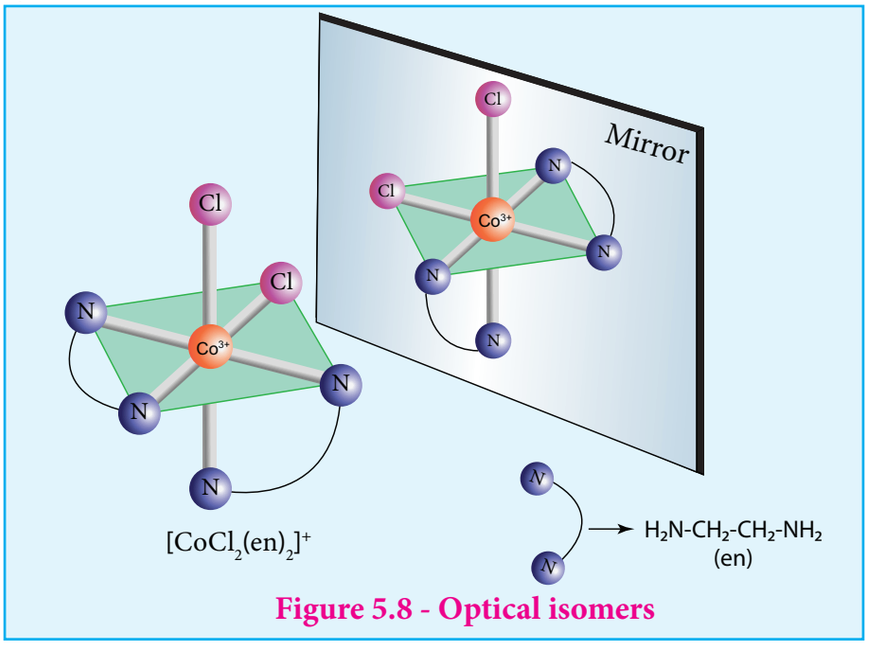


Figure 5.8 - Optical isomers

Evaluate yourself 6:

10. Draw all possible stereo isomers of a complex $Ca[Co(NH_3)Cl(Ox)_2]$

5.6 Theories of coordination compound

Alfred Werner considered the bonding in coordination compounds as the bonding between a lewis acid and a lewis base. His approach is useful in explaining some of the observed properties of coordination compounds. However, properties such as colour, magnetic property etc.. of complexes could not be explained on the basis of his approach. Following werner theory, Linus pauling proposed the Valence Bond Theory (VBT) which assumes that the bond formed between the central metal atom and the ligand is purely covalent. Bethe and Van vleck treated the interaction between the metal ion and the ligands as electrostatic and extended the Crystal Field Theory (CFT) to explain the properties of coordination compounds. Further, Ligand field theory and Molecular orbital have been developed to explain the nature of bonding in the coordination compounds. In this porton we learn the elementry treatment of VBT and CFT to simple coordination compounds.

5.6.1 Valence Bond Theory

According to this theory, the bond formed between the central metal atom and the ligand is due to the overlap of filled ligand orbitals containing a lone pair of electron with the vacant hybrid orbitals of the central metal atom.

Main assumptions of VBT:

1. The ligand → metal bond in a coordination complex is covalent in nature. It is formed by sharing of electrons (provided by the ligands) between the central metal atom and the ligand.

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